Sagnik Ghosh

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Summary

Software Engineer with strong end-to-end ownership across backend systems, applied AI, and payment infrastructure. Experienced in architecting and delivering production-grade systems from the ground up, tackling complex engineering problems across system design, data integrity, and workflow automation, with a focus on technical depth and tangible product impact.

Education

Vellore Institute of Technology, Vellore **Sep 2020 - May 2024**

Bachelor of Technology in Computer Science Engineering

- Gold Medalist - Highest scoring student in stream. CGPA - 9.40

Experience

Allo Health, Bangalore, India **Jun 2026 - Present**

Software Engineer

- Developed and shipped the core of an in-house pharmacy inventory and billing system in ~8 weeks (NestJS, TypeScript, MySQL), replacing a paid Zoho Inventory dependency that rate-limited stock lookups and held all batch data outside company control.
- Tracked stock ownership separately from physical location, letting the same batch of medicine be owned and valued differently from wholesaler to franchise pharmacy to patient - a multi-entity model the previous tool could not represent.
- Moved all pricing and tax logic server-side (discount rules, GST by product and date, hard cap at printed MRP), validated by end-to-end tests covering full B2B ad well as B2C journeys that caught 5 defects unit tests had missed.

ZenTrades.ai, Pune **Feb 2024 - May 2026**

Software Engineer **(Jul 2024 - May 2026)**

- Owned, end-to-end, a self-hosted RAG system (Django + ChromaDB vector store) that generates fire-inspection deficiency quotes and line-item recommendations grounded in NFPA reference codes, automating a manual, code-lookup-heavy step in the North American inspection workflow.
- Achieved 87% accuracy on generated deficiencies, measured against real technician feedback captured from the field, by focusing retrieval and grounding work on reducing hallucination in a compliance-sensitive domain.
- Designed and shipped, solo, a recurring-invoicing engine automating billing on a platform that has processed 1M+ invoices, using a parent-to-child template hierarchy supporting edits, pausing, date-skipping, and cancellation, with a React workflow for single-instance or bulk future-schedule edits.
- Implemented configurable, country-specific surcharge logic in Stripe, capping Canadian surcharges at the government-mandated 2.4% to ensure legal compliance and remove regulatory and churn risk ahead of market entry.
- Engineered parallel document generation using AWS Lambda and Google Apps Script, reducing generation time to under 5 minutes, eliminating API timeouts, and directly enabling expansion into the Canadian market.
- Optimized PDF generation by refactoring Puppeteer to a single shared browser instance, cutting memory usage by 50-80% and CPU load by 30-60%.
- Resolved 100+ daily production crashes to 1-2/week: implemented trace IDs, Slack crash alerts, and fixed ORM connection timeout root cause

Associate Developer **(Feb 2024 - Jul 2024)**

- Created a saved filter preferences feature allowing users to persist complex filter configurations under a single title, reducing user interaction steps by over 75%.

Skills

- **Languages:** Java, JavaScript, TypeScript, Python
- **AI/ML:** RAG pipelines, LangChain, ChromaDB, vector search, LLM integration
- **Frontend:** React.js, Next.js, React Native, HTML, CSS, Sass/SCSS, Tailwind CSS
- **Backend:** Node.js, Express.js, Django, NestJS
- **Databases:** MySQL, MongoDB, Redis
- **Tools:** Docker, Git, AWS, Linux, ElasticSearch, Nginx, TypeORM, Jest, React Testing Library

Projects

Ek Melon – AI Watermelon Quality Scanner (App Store)

- Sole developer of an iOS app using React Native, Supabase, and Gemini Vision to assess watermelon ripeness across four visual dimensions, with confidence-scored results, feedback-driven data collection, and support for 4+ Indian languages.

AI PR Review Agent

- Built an internal automated code review agent using Qwen 2.5 Coder 14B deployed on GCP Serverless Compute, replacing RovoDev, eliminating its licensing cost entirely.
- Agent analyzes pull request diffs and delivers structured, context-aware review feedback used by the engineering team on every PR.